

LIFE-CRAFT: A Multi-Agentic Conversational RAG Framework for Lifestyle Medicine Coaching with Context Traceability and Case-Based Evidence Synthesis

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Abstract. Lifestyle medicine offers a proactive and behavior-centric approach to managing chronic diseases, yet scalable, personalized coaching remains a significant challenge. We present LIFE-CRAFT (Lifestyle Intervention Facilitation Engine-Conversational RAG Framework with Traceability), a novel multi-agentic framework that leverages Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) for personalized, lifestyle medicine related guidance. Central to LIFE-CRAFT is an orchestrator agent that interprets user queries and dynamically routes them to specialized supporting agents with expertise in prominent subdomains of lifestyle medicine. LIFE-CRAFT distinguishes itself from prior LLM-based assistants by introducing adaptive RAG pipeline with specialized sub-domain expert agents alongside dynamic domain routing and iterative feedback-based self-correcting refinement mechanisms for fine-grained context traceability as each agent’s response is grounded in semantically relevant document chunks retrieved from a well-curated, expert-reviewed PDF corpus of over 7,000 PubMed articles. For every generated response, the system logs the source document context that contributed to the response, providing transparency and explainability crucial for health-related decision support. The framework is evaluated across 864 GPT-4-generated lifestyle case scenarios using both LLM-based and human-in-the-loop evaluation. Initial results indicate that LIFE-CRAFT achieves higher context alignment and perceived trustworthiness, owing to its modular architecture, agentic decision-making, and transparent evidence grounding. This work demonstrates the feasibility and promise of multi-agentic, traceable RAG systems in supporting scalable, explainable, and personalized digital lifestyle medicine coaching—laying the foundation for future health AI systems that blend reasoning, retrieval, and real-world relevance. We intend to release the code in a GitHub repository upon acceptance of the paper to promote transparency and reproducibility.

Keywords: Lifestyle Medicine, Large Language Models, RAG, Explainable AI, Multi-Agent Systems, Digital Coaching, Traceability

1 Introduction

The rising global prevalence of lifestyle-related health conditions such as obesity, diabetes, cardiovascular disease, and mental health disorders has led to increased demand for accessible, accurate, and personalized health information. However, access to expert guidance in lifestyle medicine remains limited due to various socioeconomic and

psychological barriers. Individuals often hesitate to seek in-person consultations due to stigma, cost, time constraints, or lack of immediate availability of trained professionals. In this context, the intersection of lifestyle medicine and intelligent, AI-powered tools may serve as a transformative opportunity to bridge a critical gap in care delivery and self-guided wellness management by creating intelligent, context-aware systems for supporting health behavior change, chronic disease prevention, and patient education.

We present a novel multi-agent RAG-framework specifically tailored for the lifestyle medicine domain. While prior research has explored the use of RAG technique for knowledge-grounded response generation [1,2] and multi-agent frameworks[3,4] for complex reasoning and task decomposition in domains such as drug discovery[5,6], scientific research [7-9] and customer support [10], the application of these approaches to lifestyle medicine remains largely unexplored. Despite increasing interest in digital health, most LLM-based applications have focused on clinical diagnosis, medical Q&A, or decision support, with limited attention to **preventive, lifestyle-centric interventions**. This gap is especially significant given the rising burden of lifestyle-related conditions such as obesity, diabetes, and cardiovascular diseases. Our system addresses this need by acting as a **virtual lifestyle coach**, capable of delivering personalized, timely, and evidence-backed guidance grounded in behavioral science and lifestyle medicine literature. Unlike in-person consultations with physicians or health coaches, which are often limited by time, cost, and access barriers, our virtual agent is always available, scalable, and capable of offering longitudinal support across multiple health domains. Furthermore, human lifestyle coaches may rely on subjective memory or inconsistent tracking. In contrast to general-purpose models such as GPT-4 or DeepSeek, which often rely on broad and sometimes outdated internet knowledge, our pipeline operates on a curated private repository of over 7,000 lifestyle medicine research PDFs, ensuring that responses are grounded in peer-reviewed, domain-specific evidence.

1.1 Key Innovations and Motivation:

- **Domain-Aware Agentic Architecture:** Our system uses a dynamic agent routing mechanism that directs queries to the most relevant lifestyle medicine subdomain expert agents.
- **Reduced Hallucination and Enhanced Credibility:** By confining information retrieval to a trusted internal corpus, our system mitigates hallucinations, a common issue in general-purpose LLMs.
- **Traceability and Evidence Linking:** Each generated answer includes source tracing, enhancing transparency and user trust.
- **Iterative Refinement:** The framework supports multi-turn refinement ensuring deeper coverage and tailored insights, unlike single-pass models.
- **Human and Automated Evaluation:** We employ a dual-evaluation scheme with automated visual insights for LLM response verification and human-in-the-loop evaluations.

2 Related Work:

Recent advances in retrieval-augmented generation (RAG) have significantly improved the performance of large language models on knowledge-intensive tasks. Lewis et al. [11] introduced the foundational RAG architecture by combining dense retrieval with generative modeling, enabling more factual and grounded responses in open-domain question answering. Subsequent works [12,13] have improved response quality through improved document fusion and advanced knowledge distillation techniques. Despite their effectiveness, these models typically operate in a monolithic fashion, relying on a single retrieval pipeline and lacking contextual adaptation or iterative refinement. Our system builds upon the RAG paradigm but introduces a multi-agent orchestration mechanism, enabling query-specific agent invocation, evidence aggregation, and self-corrective reasoning, all of which are particularly beneficial for complex, multi-faceted domains like lifestyle medicine.

Parallel to advancements in RAG, there has been growing interest in multi-agent LLM systems. Wu et al. [14] proposed AutoGPT, an early implementation of autonomous LLM agents capable of goal-driven task execution through tool use and memory. AgentVerse [15] extended this paradigm by facilitating multi-agent collaboration, with agents specialized in different roles. While these systems offer promising results in open-ended domains, they often lack domain-specific knowledge grounding, evidence visualization, or transparency mechanisms, features essential for trust-sensitive domains like healthcare. Our system advances this line of work by embedding agentic reasoning within a transparent, domain-constrained RAG framework tailored for preventive healthcare.

In the healthcare domain, most LLM applications are predominantly focused on clinical text summarization, medical Q&A, diagnostic support, and biomedical literature analysis. BioGPT [16], Med-PaLM [17], and ClinicalT5 [18] represent efforts to fine-tune or pretrain LLMs on biomedical corpora. However, these models are typically clinician-oriented and emphasize diagnostic accuracy over behavioral guidance. Within lifestyle medicine, digital health interventions have mostly centered around mobile health apps or behavior tracking tools [19,20], often devoid of natural language reasoning or knowledge synthesis capabilities. To the best of our knowledge, no prior work has proposed a multi-agent, RAG-based architecture particularly within the lifestyle medicine domain, integrating features such as iterative refinement, contextual evidence visualization, and domain-specific knowledge routing. Our system addresses this gap by providing an always-available, evidence-backed virtual lifestyle coach, capable of personalized, behavior-oriented responses grounded in lifestyle science.

3 Methods

The LIFE-CRAFT framework is designed to address key limitations of monolithic LLM applications by introducing modularity, traceability, and expert domain routing. We expect it to pave the way for AI to act not just as an informant, but as a domain-

guided, explainable advisor in preventive and lifestyle medicine within the growing digital health landscape.

3.1 LIFE-CRAFT Framework Architecture:

We propose and implement LIFE-CRAFT as a multi-agent RAG framework tailored for lifestyle medicine applications. The overall architecture is presented in Fig. 1. Users interact with the system via natural language queries through the user interface, which may often be in the form of detailed and potentially ambiguous case scenarios. To manage these inputs effectively, our architecture begins with an orchestrator agent, referred to as the Lifestyle Medicine Expert Coach. This agent receives the initial query, performs preliminary analysis, and decomposes the query to align it with relevant document retrieval requirements. This preprocessing step is critical for mitigating hallucinations caused by unstructured or verbose inputs.

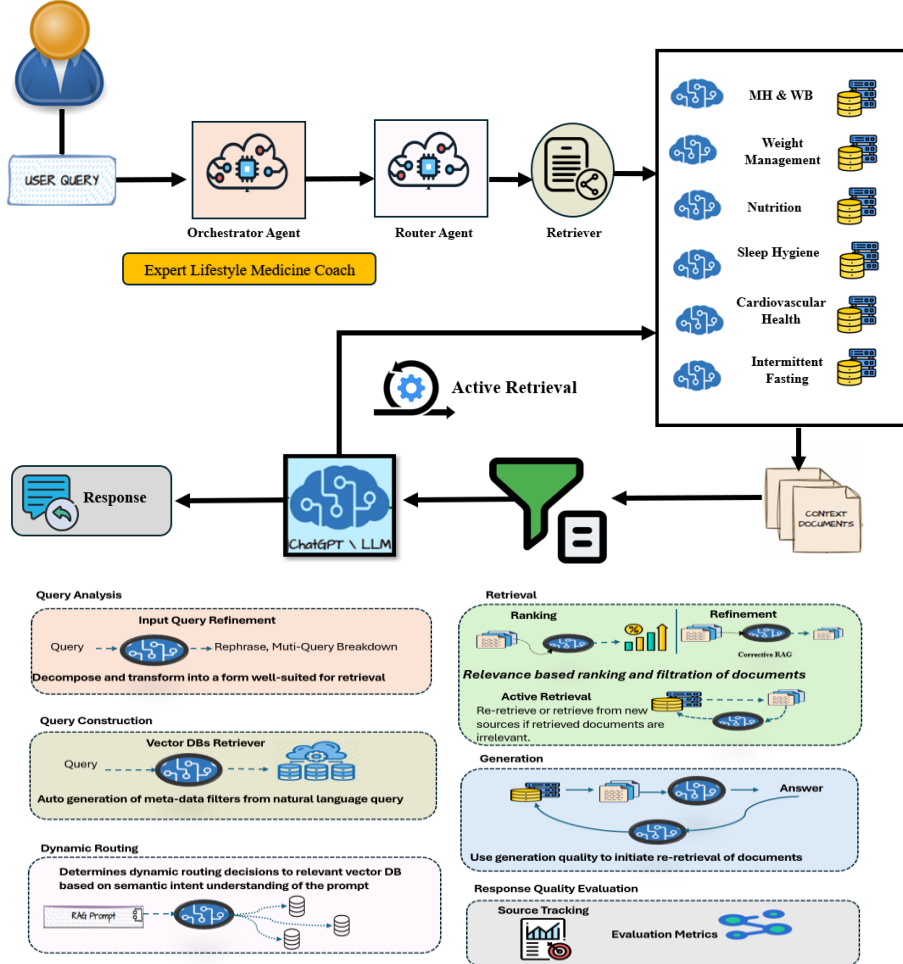


Fig. 1. The framework architecture for LIFE-CRAFT with color coded module summarization boxes.

The transformed query is then forwarded to a router agent, which based on semantic understanding of the query, performs dynamic domain-specific routing - a key innovation in our framework. This predictive routing yields reduced vector search overhead, faster multi-domain query resolution, and improved chunk relevance scoring.

Our lifestyle medicine corpus is categorized into six thematic domains. Each domain is managed by a dedicated specialist agent. The router agent can dynamically invoke one or more of these agents based on query's semantics. Before the specialist agents retrieve documents, the system enters a Query Structuring Phase, where the natural language input is translated into domain-specific query formats to facilitate metadata-based filtering and enhance semantic search for optimizing chunk retrieval precision.

To further refine the response quality, we incorporate a Corrective RAG Pipeline that includes relevance-based ranking and filtering of retrieved chunks. In cases of ambiguity or low confidence matches, the system initiates additional query transformation and may engage external knowledge sources to supplement context before generation.

Once a response is generated it is passed onto the evaluation module where responses after going through LLM assessment are presented to the user in the form of interactive insights and visualizations.

This engineered RAG workflow represents a significant advancement over traditional RAG chains, offering a more robust and intuitive approach for high-quality, context-aware response generation in lifestyle medicine scenarios. The agentic framework alongside dynamic routing, iterative refinement, corrective RAG mechanisms and evidence-based traceability with source tracking within the architecture pipeline are the key highlights that contribute to the novelty of the proposed architecture.

3.2 Knowledge Base Curation Process

A key contribution of our work is the development of an augmented lifestyle medicine database that exemplifies AI-human synergy. The PDF Dataset Curation Process for our lifestyle medicine RAG application involved a sophisticated, iterative strategy that combined human expertise, generative AI, and citation-driven expansion. The initial database curation focused on lifestyle medicine articles downloaded from PubMed via careful screening by human experts. This Initial corpus was used for prominent theme identification and topic modelling. Six core thematic areas were identified /selected through the approach for better organization and agent specialization.

We then leveraged GPT-4's generative capabilities to model diverse user scenarios, reflecting varying complexity and literature depth across identified themes by generating case scenario based synthetic QA pairs to simulate real-world queries. Next, we took these GPT generated case scenarios and ran them through Perplexity AI for citation driven expansion. What sets this work apart is the two-pass enrichment model that led to the incorporation of 7,484 highly relevant publications into the knowledge corpus. This curation process created a robust foundation for our multi-agent RAG framework, enabling nuanced, evidence-based response generation and decision-making aligned with lifestyle medicine's interdisciplinary nature.

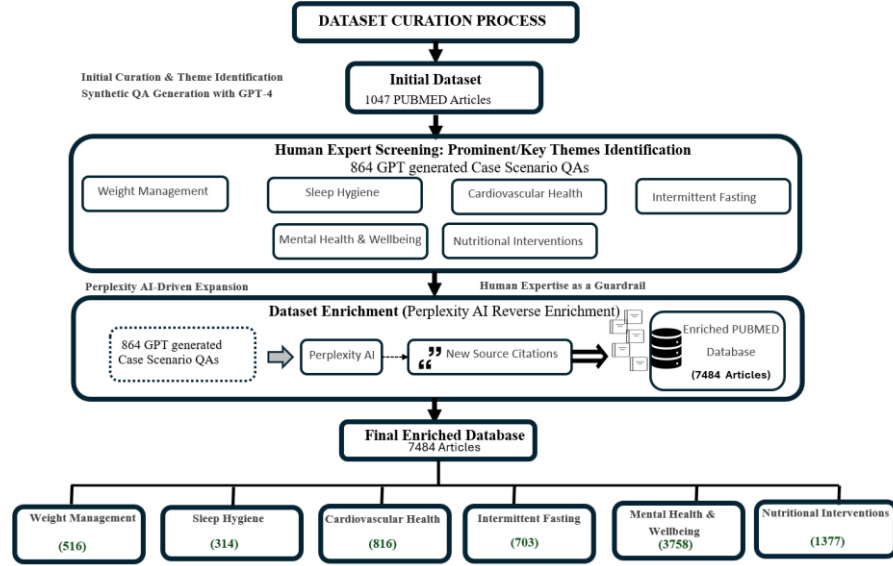


Fig. 2. Knowledge base curation strategy for RAG enabled response generation

4 Evaluation and Results:

We have created a simple user interface for query input and generated response display along with evaluation statistics display. Fig. 3 illustrates the hybrid evaluation strategy used in our framework. A user prompt is processed through the agentic response generation pipeline, producing a response that is presented to the user and simultaneously passed onto the LLM evaluation module. This module provides insights into the response quality, transparency, and explainability using visualizations for domain coverage, diversity, source tracking via chunk relevance, gap detection, and refinement.

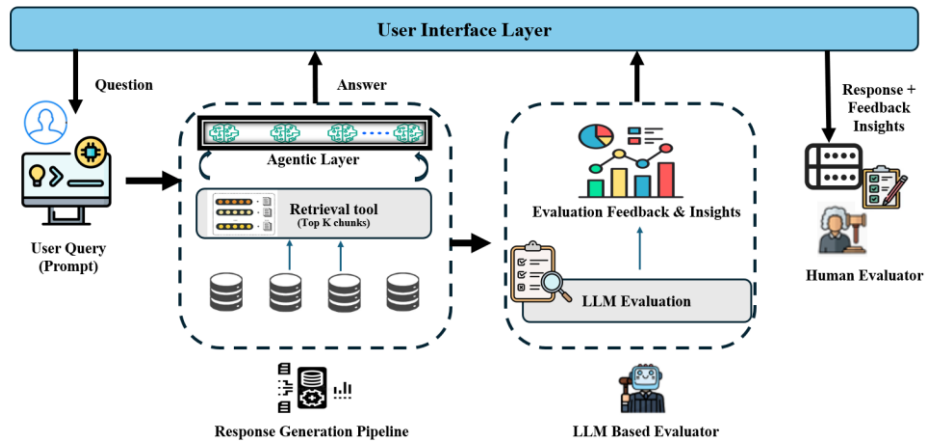


Fig. 3. Dual-evaluation strategy for quality assessment of the generated responses

The metrics used for LLM based evaluation are shown in **Table.1** and **Fig. 5** shows a sample visual feedback interface regarding iteration refinement process run for a user query giving insights regarding the gaps identified and addressed in the initial generated response for response improvisation. For human-in-the-loop evaluation, two independent experts rated the three AI-generated responses (R1: GPT-4, R2: Perplexity AI, R3: LIFE-CRAFT) per question for a sampled subset of the case scenario QA database using a five-point rubric across five dimensional criteria i.e. relevance, clinical accuracy, reasoning depth, completeness, and contextual responsiveness. Each dimension per response was scored from 1 to 5 (max total: 25). **Fig. 4** presents the evaluation results .

Table 1. LLM based evaluation metrics for assessment of generated responses

Metric	Purpose	Formula	Interpretation
Completeness (C)	Measures coverage of key concepts in retrieved evidence	$C = \frac{\sum_{i=1}^n 1(k_i \in \text{RetrievedChunks})}{n} \times 100\%$	Higher values indicate better conceptual coverage
Confidence (Conf)	Quantifies semantic alignment between query and retrieved content	$\text{Conf} = \frac{1}{m} \sum_{j=1}^m \text{sim}(q, c_j) \times 100\%$	Reflects retrieval relevance and LLM trust level
Impact (I)	Reflects value of response based on importance or actionability	$I = \frac{1}{k} \sum_{i=1}^k w_i \cdot r_i \text{ (normalized to a 5-point scale)}$	Assesses usefulness for health decision-making
Diversity (D)	Measures semantic spread among retrieved chunks	$D = \left(1 - \frac{1}{m(m-1)} \sum_{i=1}^m \sum_{j=i+1}^m \text{sim}(c_i, c_j)\right) \times 100\%$	Higher values indicate richer, non-redundant evidence

We employed Cohen’s kappa coefficient to assess inter-rater agreement, and the averaged scores are presented in the accompanying plot (Fig. 4). The human evaluation results demonstrate that our approach yields higher response quality compared to GPT and Perplexity, particularly in three key dimensions: Clinical Accuracy and Evidence Support, Depth of Reasoning and Insights, and Agentic Responsiveness and Contextual Sensitivity. These findings highlight the effectiveness of incorporating agentic retrieval-augmented generation (RAG) and explainability mechanisms in enhancing model performance.

5 Conclusion & Future Work

In this paper we propose a novel framework designed at the intersection of powerful AI tools and lifestyle medicine domain that may serve as a virtual coach. As a part of our future work, we aim to integrate real-time data from wearable devices (e.g., fitness trackers, sleep monitors, and heart rate sensors) , which has shown promise in capturing real-time health metrics [21], to further enrich the contextual understanding of user behaviors and physiological states. This will allow our system to generate even more personalized and context-sensitive responses overcoming a common limitation in

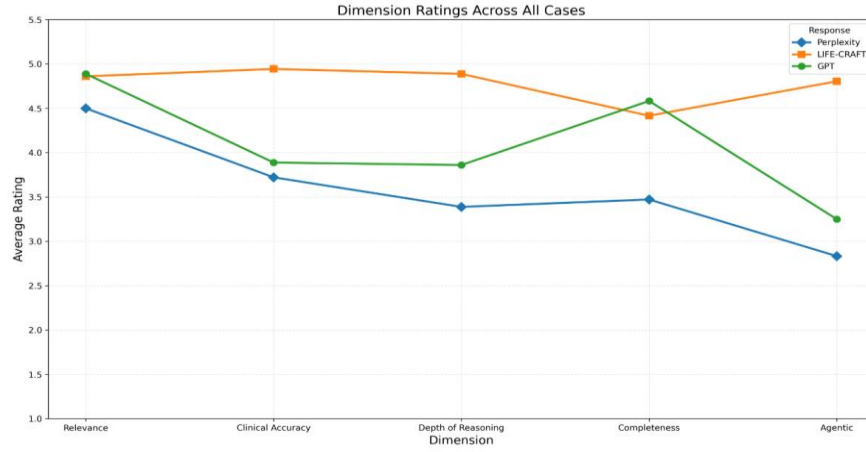


Fig. 4. Human evaluation averaged scores summarized across five dimensions

traditional consultations, where users may forget or misreport relevant lifestyle data. By seamlessly combining passive sensing with domain-aware reasoning, our platform has the potential to become an indispensable tool in preventive healthcare and self-managed wellbeing. Also, we intend to do more extensive evaluation while further enriching the data corpus across several lifestyle medicine sub domains for scaling and improving our initial framework design. We are also working on a comprehensive human centric behavioral ontology framework to characterize personas and engineer the context of the LLMs for personalized responses.



Fig. 5. Iterative refinement process results visualization tab on the user interface.

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